



A computer has learnt quantum physics

An international team of researchers are training a neural network to predict the outcomes of quantum experiments <http://digit.in/nnquant>



New kind of star predicted

A scientist has predicted a strange new kind of dense star, an object that is somewhere between a neutron star and a black hole <http://digit.in/newstar>

1970s by Nobel laureate Philip Warren Anderson. There was evidence to suggest that the quantum spin liquid state existed in some minerals, including Herbertsmithite. However, finding direct evidence for the same in a lab environment proved difficult. It was only in 2016, using a particle accelerator, that the existence of a special variant of a quantum spin liquid was found. Quantum spin liquids come in many forms with peculiar characteristics that can be of use in future quantum computers.

3D QUANTUM LIQUID CRYSTALS

Liquid crystals are themselves an exotic state of matter. The material shares characteristics with both solid crystals as well as liquids. The molecules in the liquid flow like any other regular liquid, but are all oriented in the same direction, similar to crystals. The liquid crystal state can be found in nature, including in membranes of cells. Most of us encounter liquid crystals on an everyday basis in liquid crystal displays. 3D quantum liquid crystals are quantum analogues of liquid crystals. In a quantum liquid crystal, the electrons behave like the molecules in a liquid crystal. The electrons move freely, but prefer a particular direction to flow in. A Caltech researcher came across the first observed 3D quantum liquid crystals while using lasers to study the atomic structure of rhenium compounds. The patterns observed matched with theoretical predictions of quantum liquid crystals. The find was only the first in a new class of quantum liquid crystals, and just like quantum spin liquids, this discovery could have future implications for quantum computing.

TIME CRYSTALS

Ordinary crystals have repeated patterns in space, but are constant in time. Time crystals form repeated patterns in space as well as time, and may just be the first form to be identified in a whole new category of non-equilibrium materials. A time crystal cannot exist in thermal equilibrium. Nobel

laureate Frank Wilczek first proposed the existence of non-equilibrium matter in 2012. It was in 2016 that researchers proved that a time crystal could be created in lab environments. In 2017, Norman Yao, an assistant professor in the University of California, Berkeley published a paper containing the details of creating such a crystal, and observing its properties. The time crystal was created using a chain of 10 ytterbium ions. Lasers were used to create magnetic fields and partially flip the spins of some atoms. The atoms in the chain interacted in a repetitive manner into a regular pattern that repeated in time, forming the time crystal. Think of it as a chunk of jelly that jiggles at its own period no matter how it is disturbed.

SUPERSOLID

This is a state of matter that has seemingly contradictory characteristics. A supersolid has a definite shape, or a rigid structure, just like any other solid. Like a superfluid, it can also flow without viscosity. A superfluid can escape from its container by crawling up the sides. A cup of superfluid, when stirred, would continue to spin forever. Using laser beams and vacuum chambers, researchers created the first supersolid from a superfluid in 2017. Wolfgang Ketterle had won the 2001 Nobel prize in physics for co-discovering the Bose-Einstein condensate. A team led by Ketterle worked towards creating a supersolid, by manipulating the superfluid Bose-Einstein condensate through a system of lasers. Around the same time, another group from Switzerland used a system of mirrors to create a supersolid. At the present, supersolids can only exist in the vacuum chambers at extremely low temperatures. However, the team is continuing to study the properties of the new state of matter.

GLASS

Glass is another example of an everyday material having extraordinary properties. Under a microscope, crystals have regular grid of repeated

cells. Glass however, is a disordered, amorphous material, appearing like random grains of sand, or like the grid of cells has been warped and twisted. The properties of crystals can be calculated easily, and researchers can predict how they absorb heat or what happens when they break. The same cannot be done for glass. Researchers have long suspected that a mysterious phase transition exists in glass, that exists at low temperatures. Using just a pen and paper, a postdoctoral fellow of Duke University, Shoji Yaida, solved the problem. His calculations with a pen and paper showed that at low temperatures, glass may exist at a new state of matter. Once the glass is assumed to exist in a universe with infinite dimensions, calculating the properties become as easy as calculating the properties of crystals in three dimensions. The discovery has applications in material sciences.

EXCITONIUM

In the 1960s, BI Halperin and TM Rice theorised that a new state of matter could exist under certain conditions in semiconductors. For over 50 years, researchers have tried to find or demonstrate the existence of the state of matter known as excitonium. In 2017, a team of researchers from the University of Illinois used a technique they developed called momentum-resolved electron energy-loss spectroscopy to demonstrate the existence of excitonium. Excitonium is a condensate that exhibits behavior similar to a superconductor, superfluid or an insulating electronic crystal. The state of matter is formed by a combination of escaped electrons, and the holes they leave behind. In areas of semiconductors crowded with electrons, the electrons jump out. The holes they leave behind behave like particles due to a crowd of escaped electrons around them. The holes have a positive charge, and attract the electrons, which jump in and form a particle, known as an exciton. At this point, the discovery of excitonium is fundamental research, with no known practical applications. 